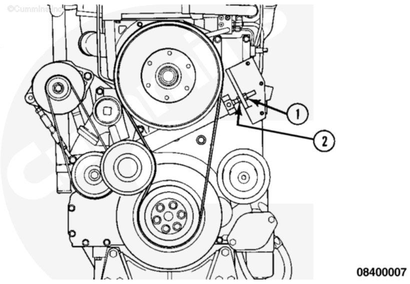
Trainings

Remove

⚠ **WARNING** ⚠

The fan belt idler is under tension. To reduce the possibility of personal injury do not place hands between the idler and the fan belt or the fan hub.



Three types of fan belt tensioning arrangements are used on the K19 engines:

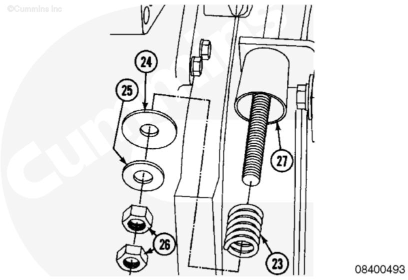
- Enclosed spring
- Control rod with spring (turnbuckle).
- Shock absorber

For the enclosed spring style of fan belt tensioner, turn the nut (1) **counterclockwise** to the end of the threaded rod, to relieve fan belt tension.

Loosen nut (2).

Note : The parts are not removed, the graphic is for clarity.

For the control rod with spring style, loosen the two jam nuts (26) to relieve the fan belt tension.



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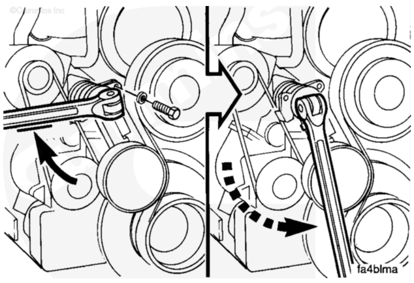
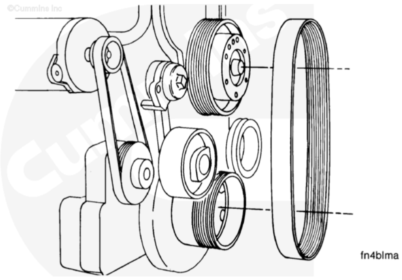


For the shock absorber style, use an 8-point socket and breaker bar or a large wrench on the lug on the idler cap to turn the arm against the spring tension. Remove the capscrew.

Slowly turn the breaker bar or wrench until the tension is relieved.

Remove the fan belt.

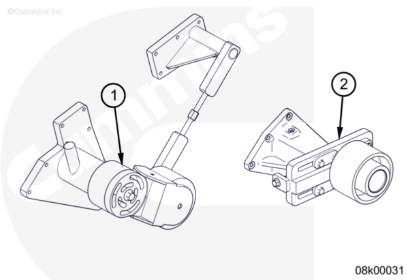


Inspect for Reuse

Inspect the tensioner assembly (1) and idler pulley (2) for any cracks or excessive wear.

If the parts are cracked or worn, they **must** be replaced.

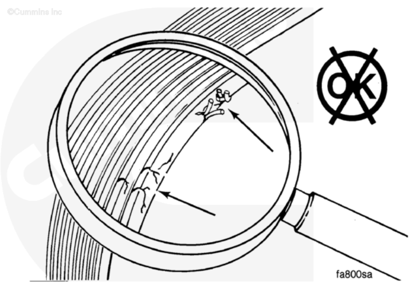
- Make sure the idler pulley moves freely.
- Make sure bearing operation is smooth and does **not** have any free play.
- Inspect the idler pulley for damage or wear.



Inspect the fan belt for:

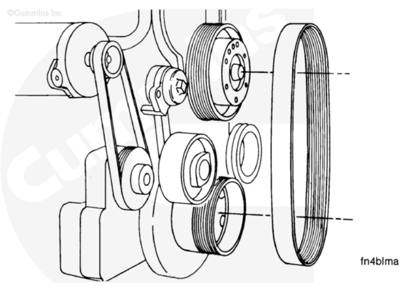
- Cracks
- Glazing
- Tears or cuts.

The fan belt **must** be replaced if damaged.

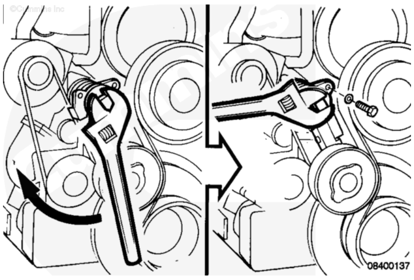


Install

Install the fan belt.



The shock absorber style of belt tensioner does **not** have an adjustment.



For the shock absorber style, use an 8-point socket and a breaker bar or a large wrench on the lug on t he idler cap to rotate the idler against the spring tension until the capscrew holes are aligned.

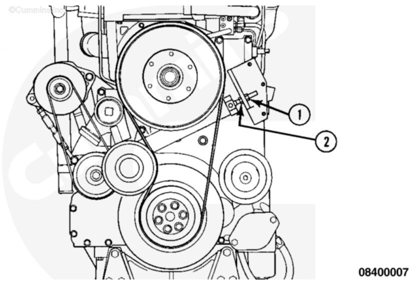
Install and tighten the capscrew.

Torque Value: 45 n•m [33 ft-lb]

Slowly turn the wrench until the idler is against the belt.

Adjust

For the enclosed spring style of belt tensioner, pull the belt tensioner until the idler pulley contacts the fan belt.



Tighten nut (1) finger-tight.

Use a wrench to tighten nut (1) nine revolutions.

Tighten nut (2).

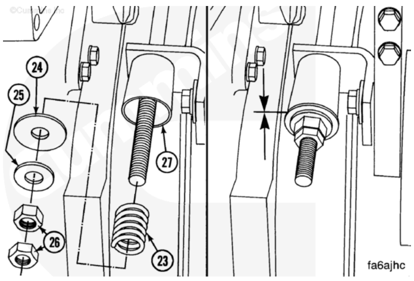
Torque Value: 81 n•m [60 ft-lb]

If the fan belt was replaced, operate the engine for 10 minutes at high idle.

Adjust the belt tension.

⚠ CAUTION ⚠

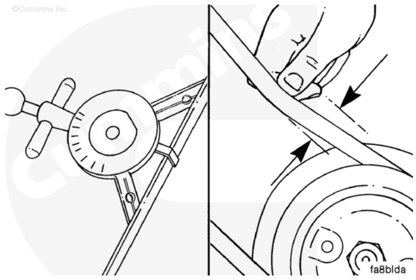
Do not tighten the inner jam nut more than one revolution after the spring retainer contacts the cup or the cup will be bent allowing a loose belt. Tighten the inner jam nut until the washer (24) contacts the cup on the bracket. Use two wrenches and tighten the outer jam nut to the inner jam nut.



For the control rod with spring belt tensioner, tighten the inner jam nut until the washer (24) contacts the cup on the bracket.

Use two wrenches and tighten the outer jam nut to the inner jam nut.

Maintenance Check



Measure the belt tension in the center span of the pulleys.

Refer to the Procedure Refer to Procedure 018-005 (/qs3/pubsys2/xml/en/procedures/99/99-018-005.html) in Section V.

An alternate method (deflection method) can be used to check belt tension by applying 110 N [25 lbf] force between the pulleys on V-belts. If the deflection is more than one belt thickness per foot of pulley center distance, the belt tension **must** be adjusted.

The tension of the fan belt does **not** need measured. The spring loaded idler used on this design maintains the correct belt tension. Refer to Procedure 100-002 (/qs3/pubsys2/xml/en/procedures/18/18-100-002.html) in Section E for location of this component.

Last Modified: 28-Mar-2017